

# ELENA MATYUSHENKO

## Senior Generative AI Engineer

business@nextry.app | [linkedin.com/in/elena-matyushenko-366526257](https://linkedin.com/in/elena-matyushenko-366526257) | [github.com/NexTryApp](https://github.com/NexTryApp) | elena.nextry.app

**Civitai: 121 published models | 111K+ downloads | 13K+ likes | 1,300+ followers**

---

## PROFESSIONAL SUMMARY

Senior Generative AI engineer building AI-powered products end-to-end — from training diffusion models and optimizing inference pipelines to shipping full-stack web platforms with real users. Led a distributed team of 8 developers (through Jan 2026) and now run production solo with AI-augmented engineering. Built and shipped NexTry (AI virtual try-on platform, 3,100+ users, 16+ generator tools, custom try-on model and a text-to-image diffusion transformer trained from scratch), BotBazar (AI bot marketplace, in beta), and TaskPilot (open-source AI agent framework). Published 121 models on Civitai with 111K+ total downloads and 1,300+ followers.

---

## CORE SKILLS

### Generative AI — Model Training & Fine-tuning

**Diffusion stacks:** SD 1.5/2.1/3, SDXL, FLUX, FLEX, HiDream, Qwen

**Fine-tuning methods:** LoRA / LyCORIS (LoHa, LoKr, LoCon), DreamBooth, Textual Inversion

**Training specializations:** style transfer, identity/character consistency, product realism, photorealism

**Training infrastructure:** custom pipelines in Python/PyTorch/Diffusers — dataloading, preprocessing, training loops, validation, hyperparameter search, logging

**Dataset engineering:** industrial web scraping across ~30 sources (anti-bot / TLS-fingerprint bypass via curl\_cffi, nodriver, yt-dlp); automated VTON triplet mining (SigLIP + YOLO11-pose + ArcFace); deduplication, quality filtering, and unified VLM-as-judge re-captioning (Qwen2.5-VL); streaming / mmap pipelines for datasets larger than RAM

**Foundation model training:** text-to-image diffusion transformer (DiT, ~202M params) trained from scratch to 470K+ steps at 512px — custom VAE latent space + Qwen2-7B text encoder + DiT denoiser; single-stream DiT architecture; edit/try-on variant with packshot reference tokens

**VAE autoencoders:** training and quality analysis of image autoencoders; identified the current 8-channel VAE quality ceiling and designed a 16-channel upgrade path (GSC skip-connections, sharpness-based data filtering, controlled GAN/adversarial component)

### Computer Vision — Control & Pipelines

**ComfyUI:** production-ready workflows, sampler/scheduler cross-testing, 2–5x inference speedups

**Control adapters:** ControlNet, IP-Adapter, PuLID

**Segmentation/detection:** SAM (Segment Anything), YOLO

**Image processing:** inpainting/outpainting, img2img, background replacement, product placement

**Video:** Luma, Kling, AnimateDiff, Runway, Hunyuan Video, WAN

### Full-Stack Development

**Frontend:** Next.js 14/15 (App Router), React 19, TypeScript, Tailwind CSS, SCSS, Radix UI / shadcn, GSAP

**Backend:** NestJS, Node.js, Express, REST API design, Swagger/OpenAPI

**Database:** PostgreSQL, Prisma ORM, SQLite, Redis

**Auth:** Google OAuth 2.0, JWT, next-auth, email/password, API key auth, 2FA (TOTP)

**Payments:** Stripe (Checkout, Connect, Elements), escrow system, commission logic

### AI Agent Development

**Agent frameworks:** custom TypeScript agent loop, tool registry, skill-based access control

**LLM integration:** 16+ providers (OpenAI, Anthropic, Groq, Ollama), function calling, prompt engineering

**Security:** command sandboxing, dangerous command blocking, approval workflows, encrypted settings

## Leadership & Team Management

**Team coordination:** led a distributed team of 8 developers (through Jan 2026) — task allocation, code review, multi-developer Git workflows; now solo with AI-augmented engineering

**AI-augmented engineering:** leveraging AI coding tools (Claude Code, Cursor) to accelerate development velocity

**Product management:** requirements gathering, feature prioritization, sprint planning, technical decision-making

**Process design:** code style guides, commit conventions, documentation standards, CI/CD pipelines

## Infrastructure & Deployment

**Containers:** Docker, docker-compose, multi-service orchestration

**Cloud:** AWS, Google Cloud, Azure, Lambda Labs, Scaleway

**Web servers:** Caddy (auto-TLS), Nginx

**CI/CD:** Git workflows, custom linters, architecture validation

**Optimization:** GPU inference tuning, batching, throughput/latency optimization

## Data & Tooling

**Core:** Python, PyTorch, Hugging Face Diffusers, Jupyter, NumPy, pandas, scikit-learn, SQL, Linux/Bash

## PROFESSIONAL EXPERIENCE

---

### Founder & Lead AI Engineer — NexTry / BotBazar / TaskPilot | 2024-Present

*Founder building AI products end-to-end — virtual try-on platform, AI bot marketplace, and open-source agent framework*

- NexTry — AI virtual try-on platform (3,100+ users, 16+ generator tools): built and launched a Next.js 15 + React 19 product — virtual try-on, mannequin generation, virtual photoshoots, colorization, pose transfer, object placement
- Training AI models from scratch: a custom virtual try-on diffusion model (59K-pair dataset — reference/garment/target with masks, OpenPose, SAM) and a text-to-image diffusion transformer (DiT, ~202M, 470K+ steps) with a custom VAE and Qwen2 text encoder
- Designed and optimized ComfyUI production workflows — systematic sampler/scheduler cross-testing for 2–5x speedups (mannequin 7.5s → 2.9s, colorize 14.4s → 4.3s, restore 33.3s → 4.0s)
- Led a distributed team of 8 developers through Jan 2026 (code review, release cycles, multi-developer Git workflows); now running production solo with AI-augmented engineering (Claude Code, Cursor)
- BotBazar — AI bot marketplace (beta): architected a full-stack monorepo from scratch (Next.js 14 + NestJS + PostgreSQL + Redis) with Stripe escrow (10% commission) + Stripe Connect payouts, multi-role auth (Google OAuth, 2FA/TOTP, API keys), and an LLM content-moderation pipeline
- TaskPilot — open-source (MIT) security-first AI agent framework: TypeScript agent loop with tool registry, memory, command-security layer, skill-based access control (4 tiers), 16+ LLM providers, and a Docker execution sandbox
- CardAI — AI product-card generator for marketplaces: 4 marketplace-ready cards from a single product photo in ~60s; separate NestJS/Python backend + Next.js frontend, PostgreSQL + Redis, ComfyUI generation, OpenAI + Gemini
- Delivered additional platforms: Alumni Network — a private alumni community (NestJS + Prisma + PostgreSQL, JWT auth, email/moderation/audit, Next.js frontend, Docker + Caddy TLS) — and a headless-browser flight/price aggregator scraper (Playwright, JS-render bypass, no API keys)
- Deployed all products with Docker on GPU cloud (AWS/GCP/Azure/Lambda/Scaleway); built a custom architecture linter enforcing layer rules, file-size limits, and import conventions

### Senior Generative AI Engineer — Freelance (Remote) | 2021–2024

*Commercial AI model training and production pipeline development*

- Published 121 models on Civitai (111K+ downloads, 13K+ likes, 1,300+ followers) — LoRA/LyCORIS for fashion, product visuals, portraits, style systems

- Built end-to-end ComfyUI production pipelines: virtual try-on, mannequin generation, photoshoots, colorization, pose transfer, object placement
- Designed and prepared large-scale datasets: annotation/captioning guidelines, cleaning, deduplication, balancing, metadata schema
- Implemented training pipelines in Python/PyTorch/Diffusers: dataloading, preprocessing, training loops, validation, logging, config iteration
- Deployed models and inference APIs with Docker on GPU cloud (AWS/GCP/Azure/Lambda/Scaleway), optimizing cost/performance for production

## EDUCATION

---

**PhD in Law** — analytical foundation applied to technical architecture, IP strategy, and compliance-aware product design

## LANGUAGES

---

Russian (Native) | English (Fluent)